

INDEPENDENCE-DISTRIBUTION-PRESERVING JOINT COVARIANCE STRUCTURES FOR THE TWO-GROUP PROBLEM

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Key words: Independence assumptions; matrix equations; multivariate quadratic forms; Wishart distribution.

Summary: We characterize the general joint covariance structure of matrix normal observations from two populations that preserves certain iid sampling properties of the sample means and sample covariance matrices. In particular, we characterize the general positive-definite joint covariance structure under which the two sample mean vectors are independent of the two sample covariance matrices. This covariance structure is also necessary and sufficient for multiples of the two sample covariance matrices to be distributed as multiples of independent noncentral Wishart random matrices. Our work extends the univariate results by Rogers (1980) and Stadge (1984) to the multivariate case.

1. INTRODUCTION

William Kruskal (1988) has noted, "An almost universal assumption in statistical models for repeated measurements of real-world quantities is that those measurements are independent, yet we know that such independence is fragile". Efforts to understand the effects of violating the "fragile"

assumption of independence have been ongoing for the past thirty years. It is well known that ignoring requisite independence assumptions for sampled observations may result in severe consequences. However, in some cases the usual independence structures for sampled observations may be changed without affecting the sampling distributions of certain statistics; that is, there are dependency structures for the data for which the sampling distribution of a statistic is the same as that under the assumption of iid observations. We shall refer to such covariance structures for the observations as *independence distribution preserving* (IDP). In this paper we characterize the general IDP covariance structure for the two-sample matrix-normal problem, generalizing the work of Rogers (1980) and Stadje (1984). We shall also provide conditions under which this general IDP covariance structure is positive definite.

Among the first to consider an IDP problem has been Halperin (1951), who has derived a sufficient IDP covariance structure for regression analysis. Many other researchers have since considered IDP covariance structures for various univariate and multivariate problems. Univariate IDP dependency research includes Faleschini (1956), Bhat (1962), Baldessari (1965, 1966, 1968), Gallo (1967), Srivastava (1980), Jeyaratnam (1982), Turner, Young and Marco (1986), Marco, Young and Turner (1987), and Tranquilli and Baldessari (1988). Multivariate IDP problems have been studied by Ghosh and Sinha (1980), Pavur (1987), Young, Pavur and Marco (1989), Pavur and Young (1990), Young, Seaman and Meaux (1992a), and Meaux, Young and Seaman (1994).

Our purpose is to develop a multivariate extension of the work by Rogers (1980) and Stadje (1984), who characterize a general form of the positive-definite IDP covariance structure for the univariate two-sample problem. Specifically, they have characterized a general positive-definite IDP covariance structure for which the univariate sample means are independent of the univariate sample variances and the sample variances are distributed as independent multiples of noncentral chi-square random variables.

Our generalization of the work of Rogers (1980) and Stadje (1984) applies to the case of multivariate statistics in a matrix-normal setting. Our main results (presented in section 3) characterize the general positive-definite IDP joint covariance matrix for which two multivariate sample mean vectors are independent of two sample covariance matrices, and the sample covariance matrices are distributed as independent multiples of noncentral Wishart random matrices. We also characterize a general positive-definite IDP joint covariance structure which guarantees the stronger condition of mutual independence among the sample mean vectors and sample covariance matrices without changing the noncentral Wishart sampling distribution of the latter.

Before developing our main results in section 3, we present notation and a preliminary theorem, of interest in its own right, which are utilized in the proof of our main theorems. We give a brief summary in section 4.

2. MATHEMATICAL PRELIMINARIES

In this section we define symbols and notation utilized throughout the remainder of the paper, and we present a key lemma which will be used to derive the main results in section 3. Additional lemmas may be found in the appendix. Let $X \equiv [X^1, X^2, \dots, X^n]'$ be an $n \times p$ random matrix where each X^i is a $p \times 1$ random vector. We say that X has a matrix-normal distribution with mean matrix $\mu = [\mu^1, \mu^2, \dots, \mu^n]'$ where $E(X^i) = \mu^i$ and covariance matrix V if $\text{vec}(X) \equiv [X^1, X^2, \dots, X^n]'$ has a multivariate normal distribution with mean $\text{vec}(\mu')$ and covariance matrix V . We use the notation $X \sim MN(\mu, V)$ to denote that the random matrix X has a matrix-normal distribution with mean matrix μ and covariance matrix V , and we use the notation $Y \sim N(\mu, D)$ to indicate that the $p \times 1$ random vector Y has a multivariate normal distribution. Note that $\text{vec}(X)$ is an $np \times 1$ vector, V is an $np \times np$ matrix, μ is an $n \times p$ matrix, and μ^i is a $p \times 1$ vector.

For a $p \times p$ random matrix let $W_p(k, \Sigma, \Delta)$ denote a noncentral

Wishart distribution having k degrees of freedom and noncentrality parameter Δ . A central Wishart distribution ($\Delta = 0$) having k degrees of freedom is denoted by $W_p(k, \Sigma)$. Also, we shall use the notation $\chi^2(k, \delta)$ to denote a noncentral chi-squared random variable having k degrees of freedom and noncentrality parameter δ .

Let I_k be the $k \times k$ identity matrix, let M_i be the $n_i \times n_i$ matrix of ones, let e_i be an $n_i \times 1$ vector of ones, and define the idempotent matrix $P_i \equiv I_{n_i} - \frac{1}{n_i} M_i$, $i = 1, 2$. Let X_i be an $n_i \times p$ matrix. Define $\bar{X}_i \equiv \frac{1}{n_i} X_i' e_i$ and $S_i \equiv X_i' P_i X_i = \sum_{j=1}^{n_i} (X_i^j - \bar{X})(X_i^j - \bar{X})'$ where X_i^k is the transpose of the k -th row of X_i . We dispense with the subscript in the case corresponding to a single population. The notation $A \otimes B$ denotes the Kronecker matrix product, which is defined to be $(a_{ij} \otimes B)$ as given in Anderson (1984, p. 599). We use the notation U^- to denote a generalized inverse of the matrix U and the notation U^+ to denote the Moore-Penrose pseudo-inverse of the matrix U . Finally, we indicate that a square matrix A is positive-definite with the notation $A >_{pd} 0$ and that a square matrix B is nonnegative-definite by the notation $B \geq_{nd} 0$.

We now present a theorem that is quite important in our derivations. The following theorem characterizes the general positive-definite IDP covariance structure of a matrix-normal random variable such that the sample covariance matrix is distributed as a multiple of a noncentral Wishart random matrix and is independent of the sample mean. The theorem is a multivariate extension of univariate results by Bhat (1962), Rogers (1980), and Stadje (1984).

THEOREM 1: Let $X \sim MN(\mu, \Xi)$ where μ is an $n \times p$ matrix and the $np \times np$ covariance matrix $\Xi >_{pd} 0$. Denote by X^i the transpose of the i -th row of X , $i = 1, 2, \dots, n$. Suppose that the p -dimensional marginal distribution of each vector X^i is $N(\mu_i, \Xi_i)$. Then, $S \sim cW_p(n-1, \Sigma, \Delta)$ where $\Sigma >_{pd} 0$, $\Delta = \Sigma^{-\frac{1}{2}} \mu' P \mu \Sigma^{-\frac{1}{2}}$, and S is independent of $\bar{X}$ if and only if

$$(2.1) \quad \Xi = c \left(\mathbf{I}_n - \frac{1}{n} \mathbf{M} \right) \otimes \Sigma + \mathbf{M} \otimes \mathbf{W}$$

where $\mathbf{W} >_{pd} 0$ is arbitrary and $c > 0$.

PROOF OF THEOREM 1. Sufficiency follows from the fact that the matrix defined in (2.1) satisfies the matrix equations (A.1) and (A.2) found in the appendix. We now prove necessity. Let $\mathbf{S} \equiv \mathbf{X}' \mathbf{P} \mathbf{X} \sim c \mathbf{W}_p(n-1, \Sigma)$ where Σ is a positive-definite matrix, and let $\mathbf{S}$ be independent of $\bar{\mathbf{X}}$. Noting that $\frac{1}{n_i} \mathbf{e}_i \mathbf{e}_i' = \frac{1}{n_i} \mathbf{M}_i$, $i = 1, 2$, we may apply the fact that $\bar{\mathbf{X}} \bar{\mathbf{X}}' = \frac{1}{n} \mathbf{X}' \mathbf{M} \mathbf{X}$, as well as the fact that functions of independent statistics are independent, and also apply lemma A1 to yield

$$(2.2) \quad \begin{aligned} (\mathbf{P} \otimes \mathbf{I}_p) \Xi \left(\frac{1}{n} \mathbf{M} \otimes \mathbf{I}_p \right) &= (\mathbf{P} \otimes \mathbf{I}_p) \Xi \left[\left(\mathbf{P} - \mathbf{P} + \frac{1}{n} \mathbf{M} \right) \otimes \mathbf{I}_p \right] \\ &= (\mathbf{P} \otimes \mathbf{I}_p) \Xi (\mathbf{P} \otimes \mathbf{I}_p) - (\mathbf{P} \otimes \mathbf{I}_p) \Xi \\ &= 0. \end{aligned}$$

Using (A.2), we may rewrite equation (2.2) so that the positive-definite matrix Ξ must satisfy the matrix equation

$$(2.3) \quad (\mathbf{P} \otimes \mathbf{I}_p) \Xi = c \mathbf{P} \otimes \Sigma.$$

Application of lemma A3 to equation (2.3) and some algebraic simplification provide a general form of the nonnegative-definite solution, which is given in (2.1) with $\mathbf{W} >_{pd} 0$.

Now, let $\mathbf{G}$ be the Helmert matrix defined by

$$\mathbf{G} = \begin{bmatrix} \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \cdots & \frac{1}{\sqrt{n}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} & 0 & \cdots & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ \frac{1}{\sqrt{n(n-1)}} & \frac{1}{\sqrt{n(n-1)}} & \frac{1}{\sqrt{n(n-1)}} & \cdots & \frac{-(n-1)}{\sqrt{n(n-1)}} \end{bmatrix}$$

Then, in (2.1) $\Xi >_{pd} 0$ if and only if $(G \otimes I_p) \Xi (G' \otimes I_p) >_{pd} 0$. Since $(G \otimes I_p) \Xi (G \otimes I_p) = \text{Diag} \{nW, c\Sigma, c\Sigma, \dots, c\Sigma\}$, it follows that $\Xi >_{pd} 0$ if and only if $\Sigma >_{pd} 0$ and $W >_{pd} 0$.

The following corollary characterizes the general mean matrix and general positive-definite IDP covariance matrix such that a multiple of the sample covariance matrix is distributed as a multiple of a central Wishart distribution. A portion of this corollary is stated in theorem 3 in Pavur (1987). Our method of proof differs from a method of proof alluded to by Pavur (1987). Furthermore, we give conditions on the general form of the IDP covariance matrix that insure positive-definiteness.

COROLLARY 1. Let $X \sim MN(\mu, \Xi)$ where the $np \times np$ covariance matrix $\Xi >_{pd} 0$. Let S be defined as in theorem 1, and let the $p \times p$ matrix $\Sigma >_{pd} 0$. Then, $S \sim cW_p(n-1, \Sigma)$ and is independent of $\bar{X}$ if and only if

$$(2.1) \quad \Xi = c(I_n - \frac{1}{n}M) \otimes \Sigma + M \otimes W$$

where $W >_{pd} 0$ is arbitrary, $c > 0$, and $\mu = \frac{1}{n}MZ$ where Z is an arbitrary $n \times p$ matrix.

PROOF OF COROLLARY 1. The proof follows from theorem 1, lemma A2, and the fact that the general solution of the matrix equation $(I_n - \frac{1}{n}M)\mu = 0$ is of the form $\mu = \frac{1}{n}MZ$ where Z is an arbitrary $n \times p$ matrix.

Note that the mean matrix $\mu = \frac{1}{n}MZ$, where Z is an arbitrary $n \times p$ matrix, is more general than the mean matrix assumed in theorem 3 of Pavur (1987), which is of the form $\mu = [\nu, \nu, \dots, \nu]'$ where ν is a $p \times 1$ vector.

3. THE MAIN RESULTS

We now prove two theorems which comprise our main results. Here we consider the case in which we are sampling from two distinct groups. The first theorem characterizes the general positive-definite IDP joint covariance structure for the two-sample matrix-normal problem. This IDP covariance

structure is necessary and sufficient to not only cause the two sample means to be independent of the two sample covariance matrices, but also to cause multiples of the two sample covariance matrices to be distributed as multiples of independent noncentral Wishart random matrices. Our theorem is a multivariate extension of a univariate result proved by Rogers (1980) and Stadje (1984). The second theorem characterizes the general positive-definite IDP joint covariance structure for which the two sample mean vectors and the two sample covariance matrices are *mutually* independent while multiples of the sample covariance matrices are distributed as multiples of noncentral Wishart random matrices.

THEOREM 2. Let $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim MN(\mu, V)$ where $\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$, and let

$$(3.1) \quad V = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix} >_{pd} 0$$

where X_i has dimension $n_i \times p$, $i = 1, 2$. Then, (S_1, S_2) is independent of $(\bar{X}_1, \bar{X}_2)$, and $S_i \sim c_i W_p(n_i - 1, \Sigma_i, \Delta_i)$ where $c_i > 0$, the $p \times p$ matrix $\Sigma_i >_{pd} 0$, $\Delta_i = c_i^{-1} \Sigma_i^{-\frac{1}{2}} \mu_i' P_i \mu_i \Sigma_i^{-\frac{1}{2}}$, $i = 1, 2$, with S_1 independent of S_2 if and only if

$$V_{ii} = c_i \left(I_{n_i} - \frac{1}{n_i} M_i \right) \otimes \Sigma_i + M_i \otimes W_i,$$

and

$$V_{12} = \frac{1}{n_1 n_2} (M_1 \otimes I_p) T (M_2 \otimes I_p)$$

where $W_i >_{pd} 0$ are arbitrary and T is any $n_1 p \times n_2 p$ matrix such that $V >_{pd} 0$.

PROOF OF THEOREM 2. By theorem 1 $S_i \sim c_i W_p(n_i - 1, \Sigma_i, \Delta_i)$ and $\bar{X}_i$ is independent of S_i if and only if

$$V_{ii} = c_i \left(I_{n_i} - \frac{1}{n_i} M_i \right) \otimes \Sigma_i + M_i \otimes W_i,$$

$i = 1, 2$. Also, noting that

$$S_1 = [X_1' \ X_2'] \begin{bmatrix} P_1 & 0 \\ 0' & 0'0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \quad \text{and} \quad S_2 = [X_1' \ X_2'] \begin{bmatrix} 0'0 & 0' \\ 0 & P_2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix},$$

it follows from lemma A1 that S_1 is independent of S_2 if and only if

$$\begin{bmatrix} (P_1 \otimes I_p) & 0 \\ 0' & 0'0 \end{bmatrix} \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix} \begin{bmatrix} 0'0 & 0' \\ 0 & (P_2 \otimes I_p) \end{bmatrix} = 0,$$

which simplifies to the condition

$$(3.2) \quad (P_1 \otimes I_p) V_{12} (P_2 \otimes I_p) = 0.$$

Similarly, noting that $\frac{1}{n_i} e_i e_i' = \frac{1}{n_i} M_i$, $i = 1, 2$, we may apply the fact

that $\bar{X} \bar{X}' = \frac{1}{n} X' M X$, that lemma A1, and that functions of independent statistics are independent to show that $\bar{X}_1$ is independent of S_2 if and only if

$$(3.3) \quad (P_2 \otimes I_p) V_{12} (M_1 \otimes I_p) = 0.$$

Likewise, $\bar{X}_2$ is independent of S_1 if and only if $(P_1 \otimes I_p) V_{12} (M_2 \otimes I_p) = 0$. Using lemma A4, we see that condition (3.2) implies that

$$(3.4) \quad \begin{aligned} V_{12} &= R - (P_1 \otimes I_p)^* (P_1 \otimes I_p) R (P_2 \otimes I_p) (P_2 \otimes I_p)^* \\ &= R - (P_1 \otimes I_p) R (P_2 \otimes I_p) \end{aligned}$$

where R is arbitrary because $(P_i \otimes I_p)^* = (P_i \otimes I_p)$ for $i = 1, 2$. Note that condition (3.3) and result (3.4) imply that

$$\begin{aligned} 0 &= (M_1 \otimes I_p) [R - (P_1 \otimes I_p) R (P_2 \otimes I_p)] (P_2 \otimes I_p) \\ &= (M_1 \otimes I_p) R (P_2 \otimes I_p) - (M_1 P_1 \otimes I_p) R (P_2 \otimes I_p) \\ &= (M_1 \otimes I_p) R (P_2 \otimes I_p) \end{aligned}$$

since M_1 is orthogonal to P_1 . Thus, by lemma A4 we may express R as

$$(3.5) \quad R = U - \frac{1}{n_1}(M_1 \otimes I_p)U(P_2 \otimes I_p)$$

where U is an arbitrary $n_1 p \times n_2 p$ matrix. It follows from (3.4) and (3.5) that

$$\begin{aligned} V_{12} &= U - \frac{1}{n_1}(M_1 \otimes I_p)U(P_2 \otimes I_p) - (P_1 \otimes I_p)[U - \frac{1}{n_1}(M_1 \otimes I_p)U(P_2 \otimes I_p)](P_2 \otimes I_p) \\ &= U - \frac{1}{n_1}(M_1 \otimes I_p)U[(I_p - \frac{1}{n_2}M_2) \otimes I_p] - (P_1 \otimes I_p)U[(I_p - \frac{1}{n_1}M_2) \otimes I_p] \\ (3.6) \quad &= \frac{1}{n_2}U(M_2 \otimes I_p). \end{aligned}$$

Similarly, $(P_1 \otimes I_p)V_{12}(M_2 \otimes I_p) = 0$ implies that $(P_1 \otimes I_p)U(M_2 \otimes I_p) = 0$, which, because of lemma A4, implies that

$$(3.7) \quad U = T - \frac{1}{n_2}(P_1 \otimes I_p)T(M_2 \otimes I_p).$$

Substituting the right-hand side of (3.7) for U in (3.6) yields

$$(3.8) \quad V_{12} = \frac{1}{n_1 n_2}(M_1 \otimes I_p)T(M_2 \otimes I_p)$$

where T is free to vary over the set of $n_1 p \times n_2 p$ matrices such that $V >_{pd} 0$.

We now determine restrictions for the matrix T such that $V >_{pd} 0$. The covariance structure for the two marginal random matrices X_1 and X_2 , which preserve the required properties and are positive definite, are given in (2.1). Hence, let V_{11} and V_{22} be defined as in (2.1). Also, let V_{12} be defined as in (3.8). From lemma A5 we have that $V >_{pd} 0$ if and only if, for $i = 1, 2$, $i \neq j$,

$$(3.9) \quad V_{ii} - \frac{1}{(n_i n_j)^2} [(M_i \otimes I_p) T (M_j \otimes I_p)] V_{jj}^{-1} [(M_i \otimes I_p) T (M_j \otimes I_p)]' >_{pd} 0.$$

Hence, $V >_{pd} 0$ if and only if, given V_{11} and V_{22} , T simultaneously satisfies the pair of constraints given in (3.9).

We now prove the second of our main results. The next theorem characterizes the general form of the positive-definite IDP joint covariance structure such that the sample means and the sample covariance matrices for the two groups are mutually independent and multiples of the sample covariance matrices are distributed as multiples of Wishart random matrices.

THEOREM 3. Let $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim MN(\mu, V)$ where $\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$ and $V = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix} >_{pd} 0$. Let $\bar{X}_i$ and S_i , $i = 1, 2$, be defined as in theorem 2. Then, $\bar{X}_1, \bar{X}_2, S_1$ and S_2 are mutually independent and $S_i \sim c_i W_p(n_i - 1, \Sigma_i, \Delta_i)$ where $\Delta_i = \Sigma_i^{-\frac{1}{2}} \mu_i' P \mu_i \Sigma_i^{-\frac{1}{2}}$, $i = 1, 2$, if and only if V_{ii} , $i = 1, 2$, are defined as in theorem 2 and $V_{12} = 0$.

PROOF OF THEOREM 3. We need derive only necessary and sufficient conditions on V so that $\bar{X}_1$ and $\bar{X}_2$ are independent because the covariance structures which yield all other independence relationships and the appropriate distributions of S_i , $i = 1, 2$, have been proved in theorems 1 and 2. We first show necessity. Again, utilizing lemma A1 and the fact that functions of independent statistics are independent, we have that $\bar{X}_1$ and $\bar{X}_2$ are independent if and only if

$$\begin{bmatrix} (M_1 \otimes I_p) & 0 \\ 0' & 0' \end{bmatrix} \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix} \begin{bmatrix} 0' & 0' \\ 0' & (M_2 \otimes I_p) \end{bmatrix} = 0,$$

which, in turn, simplifies to

$$(3.10) \quad (M_1 \otimes I_p) V_{12} (M_2 \otimes I_p) = 0.$$

Substituting the right-hand side of (3.8) for V_{12} into (3.10) and simplifying yields

$$(3.11) \quad (M_1 \otimes I_p)T(M_2 \otimes I_p) = 0.$$

Thus, using lemma A4 to solve for T in (3.11), we have

$$(3.12) \quad T = F - \frac{1}{n_1 n_2} (M_1 \otimes I_p)F(M_2 \otimes I_p)$$

where F is an arbitrary $n_1 p \times n_2 p$ matrix. It follows from (3.8) and (3.12) that

$$\begin{aligned} V_{12} &= \frac{1}{n_1 n_2} (M_1 \otimes I_p) \left[F - \frac{1}{n_1 n_2} (M_1 \otimes I_p)F(M_2 \otimes I_p) \right] (M_2 \otimes I_p) \\ &= 0. \end{aligned}$$

As for sufficiency, (3.10) is obviously satisfied when $V_{12} = 0$.

The following corollary extends the results of Rogers (1980) and Stadge (1984) by giving necessary and sufficient conditions on the joint positive-definite covariance structure in order that the univariate statistics $\bar{X}_i$, S_i^2 , $i = 1, 2$, are mutually independent and S_i^2 , $i = 1, 2$, are distributed as multiples of noncentral chi-square random variables.

COROLLARY 2. Let $\begin{bmatrix} X^1 \\ X^2 \end{bmatrix} \sim N(\mu, D)$ where $\mu = \begin{bmatrix} \mu^1 \\ \mu^2 \end{bmatrix}$ and $D = \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix}$.

Let $\bar{X}_i$, S_i^2 be defined as $\bar{X}_i$ and S_i , $i = 1, 2$, respectively, with $p = 1$.

Then, for $i = 1, 2$, $\bar{X}_i$ and S_i^2 are mutually independent and $S_i^2 \sim c_i \chi^2(n_i - 1, \delta_i)$

if and only if $D_{ii} = c_i I_{n_i} + d_i M_i$ and $D_{12} = 0$ where $c_i > 0$, $d_i + c_i/n_i > 0$.

PROOF OF COROLLARY 2. The proof follows directly from theorem 3.

4. SUMMARY

In theorem 1 we have presented a characterization of the general form of the IDP covariance matrix of a matrix-normal random matrix such that a multiple of the usual multivariate sample covariance matrix is distributed as a multiple of a noncentral Wishart random matrix and is independent of the sample mean. Moreover, we give conditions on the general form of the IDP covariance matrix that insures positive-definiteness. In corollary 1 we characterize the general form of the mean matrix and the positive-definite IDP covariance matrix such that a multiple of the usual multivariate sample covariance matrix is distributed as a multiple of a central Wishart random matrix and is independent of the sample mean. Thus, corollary 1 is a complete proof of a slight generalization of a result partially stated in theorem 4 in Pavur (1987).

We have established in theorem 2 that, assuming joint matrix normality of the observations, the between-group covariance V_{12} need not be the zero matrix in order for the joint distribution of $\bar{X}_1$ and $\bar{X}_2$ to be independent of the joint distribution of S_1 and S_2 . Furthermore, it is not necessary for V_{12} to be the zero matrix in order that $S_i \sim c_i W_p(n_i - 1, \Sigma_i)$, $i = 1, 2$, and for S_1 and S_2 to be independent. Theorem 2 also provides a characterization of the positive-definite IDP joint covariance matrix yielding these conclusions. Theorem 2 is, thus, a multivariate extension of the univariate results found in the theorem in Rogers (1980) and in theorem 3 in Stadje (1984).

Theorem 3 characterizes a general form of the positive-definite IDP joint covariance matrix such that $\bar{X}_1$, $\bar{X}_2$, S_1 , and S_2 are *mutually* independent while S_1 and S_2 are distributed as multiples of noncentral Wishart random matrices.

In his discussion of the "casual assumption of independence", Kruskal (1988) has mused that if there were a moral to his theme, it would be, "Do not multiply lightly." In the case of the two-sample matrix-normal problem, we have shown the extent to which we can avoid casual invocation

of independence without losing many of the properties of the usual independence-induced properties of the statistics $\bar{X}_i$ and S_i , $i = 1, 2$.

APPENDIX

The following lemmas are used in the proof of the theorems in sections 2 and 3. The first lemma gives necessary and sufficient conditions for two multivariate quadratic forms to be independent given that the quadratic forms are generated from the same matrix-normal random matrix having a positive-definite covariance matrix. A proof of the lemma may be found in corollary 2.6.2 of Wong, Masaro, and Wang (1991).

LEMMA A1. *Let $X \sim MN(\mu, V)$ where μ is an $n \times p$ mean matrix and the $np \times np$ covariance matrix $V >_{pd} 0$. Consider the multivariate quadratic forms $X'AX$ and $X'BX$ where A and B are symmetric $n \times n$ matrices. Then, the multivariate quadratic forms $X'AX$ and $X'BX$ are independent if and only if*

$$(A.1) \quad (A \otimes I_p)V(B \otimes I_p) = 0.$$

The second lemma gives necessary and sufficient conditions on the positive-definite covariance structure such that a certain type of multivariate quadratic form generated from a matrix-normal random matrix with nonzero mean matrix is distributed as a noncentral Wishart random matrix. A proof may be found in Young, Seaman, and Meaux (1992b).

LEMMA A2. *Consider the multivariate quadratic form $X'AX$ given that $X \sim MN(\mu, V)$ where μ is an $n \times p$ mean matrix and the $np \times np$ covariance matrix $V >_{pd} 0$. Let $\Sigma >_{pd} 0$ and let A be an $n \times n$ symmetric matrix. Then, $X'AX \sim cW_p(k, \Sigma, \Delta)$ if and only if*

$$(A.2) \quad (A \otimes I_p)V(A \otimes I_p) = cA \otimes \Sigma$$

where $k = \text{rank}(A)$, $c > 0$, and $\Delta = c^{-1} \Sigma^{-\frac{1}{2}} \mu' A \mu \Sigma^{-\frac{1}{2}}$. Moreover, $X'AX \sim cW_p(k, \Sigma)$ if and only if $A\mu = 0$.

The next lemma, which has been proved in Khatri and Mitra (1976), gives a general form of the nonnegative-definite solution to a particular linear matrix equation.

LEMMA A3. Let $C^{m \times n}$ denote the linear space of matrices over the complex field, let $C_m^{\geq}$ be the cone consisting of all nonnegative-definite matrices in $C^{m \times m}$, and let A and C be matrices in $C^{m \times n}$. Also, for $K \in C^{m \times n}$ let K^* denote the conjugate transpose of K . The matrix equation $AY = C$ has a nonnegative-definite solution if and only if $CA^* \succeq_{\text{nd}} 0$ and $\text{rank}(CA^*) = \text{rank}(C)$. Provided a solution exists, the general nonnegative-definite solution is of the form

$$Y = C^*(CA^*)^-C + (I - A^-A)U(I - A^-A)^*$$

where $(CA^*)^-$ and A^- are arbitrary generalized inverses of CA^* and A , respectively, and $U \in C_n^{\geq}$.

Next, we state a lemma concerning the general solution of a certain linear matrix equation. The lemma is well known and may be found in Lewis and Odell (1971, p. 10) or Magnus and Neudecker (1988, p. 37).

LEMMA A4. Let A , B , C and Y be real matrices such that the matrix equation

$$(A.3) \quad AYB = C$$

is defined with A , B , and C known and Y unknown. Then, the matrix equation (A.3) has a solution if and only if $AA^+CB^+B = C$ and the general form of the solution is

$$Y = A^+CB^+ + F - A^+AFBB^+$$

where F is an arbitrary matrix.

The following lemma, proved in Albert (1969), gives necessary and sufficient conditions for a matrix to be nonnegative definite in terms of certain functions of submatrices.

LEMMA A5. Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

where A_{11} is a symmetric $m \times m$ matrix, A_{22} is a symmetric $n \times n$ matrix, and $A_{21} = A_{12}'$. Then, A is positive definite if and only if $A_{ii} >_{\text{pd}}$ 0, $i = 1, 2$, and $A_{ii} - A_{ij}A_{jj}^{-1}A_{ji} >_{\text{pd}}$ 0, $i, j = 1, 2$, $i \neq j$.

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